

The Transcendental Hierarchy

Extending the Integer Pair Basis to Complete Elliptic Integrals and the 4-Loop Boundary

Registry: [\[HOWL-MATH-3-2026\]](#)

Series Path: [\[HOWL-MATH-1-2026\]](#) → [\[HOWL-MATH-2-2026\]](#) → [\[HOWL-MATH-3-2026\]](#)

Physics Path: MATH-3 feeds [\[HOWL-PHYS-5-2026\]](#) and [\[HOWL-PHYS-6-2026\]](#) by extending the integer pair basis toward the 4-loop electron g-2.

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I. ABSTRACT

[\[HOWL-MATH-2-2026\]](#) established integer pair representations for 17 transcendental constants. [\[HOWL-PHYS-5-2026\]](#) and [\[HOWL-PHYS-6-2026\]](#) used five of these — π , $\ln(2)$, $\zeta(3)$, $\zeta(5)$, $\text{Li}_4(1/2)$ — to compute the electron anomalous magnetic moment through 3-loop order in exact rational arithmetic. At 4-loop, complete elliptic integrals appear and the integer pair framework stalls. This paper extends the MATH-2 basis in three directions: complete elliptic integrals at rational arguments as convergent rational series, accelerated rational series for odd zeta values, and identification of the transcendental hierarchy that governs which constants appear at which order of perturbation theory. The extended basis is applied to Laporta's numerically-known 4-loop master integrals via PSLQ integer relation detection.

II. THE CURRENT BASIS AND ITS LIMITS

2.1 The MATH-2 Basis

MATH-2 computed 17 transcendental constants as integer pairs — exact ratios of two integers from convergent rational series. Each constant is computed at a controlled precision: 100 to 999 correct digits depending on the series convergence rate. The constants include π (via Machin's formula, geometric convergence, 999 digits at 160 terms), $\ln(2)$ (via $\text{arctanh}(1/3)$, geometric convergence, 999 digits at 160 terms), e (via factorial series,

geometric convergence), $\zeta(n)$ for $n = 2, 3, 4, 5$ (via various series), and polylogarithms $\text{Li}_n(1/2)$ for small n .

An integer pair is a Fraction p/q with p, q integers, computed by truncating a convergent rational series at a specified number of terms. The truncation error is bounded by the next term in the series. The result is not an approximation in the floating-point sense — it is an exact rational number that differs from the true transcendental value by a known, bounded amount.

2.2 The Five That Appear in QED Through 3-Loop

The electron anomalous magnetic moment $a_e = \sum A_n (\alpha/\pi)^n$ has coefficients A_1, A_2, A_3 that are rational combinations of exactly five transcendental constants:

Constant	MATH-2 series	Convergence	Digits at N terms	Appears at
π	Machin (arctan)	Geometric (~1.4 bits/term)	999 at 160	2-loop (as π^2)
$\ln(2)$	$2 \cdot \text{arctanh}(1/3)$	Geometric (~1.6 bits/term)	999 at 160	2-loop
$\zeta(3)$	Central binomial	Geometric (~1.0 bit/term)	114 at 180	2-loop
$\zeta(5)$	Alternating eta	Linear ($1/N^5$)	20 at 10,000	3-loop
$\text{Li}_4(1/2)$	Direct sum $1/(2^n n^4)$	Geometric (~1 bit/term)	100 at 300	3-loop

These five arise because QED Feynman diagrams through 3-loop reduce, via Feynman parametrization and integration-by-parts identities, to integrals that evaluate to rational combinations of these constants. The reduction is complete at 3-loop: every diagram factors into products of one-loop subintegrals or produces constants from the $\zeta/\ln/\text{Li}$ family.

The origin of each constant can be traced to a specific integral structure. π arises from the Dirac trace and angular integrals. $\ln(2)$ arises from threshold integrals where the virtual particle goes on-shell at the halfway point of the Feynman parameter range. $\zeta(3)$ arises from the three-fold nested integration of $1/n$ -type denominators. $\zeta(5)$ and $\text{Li}_4(1/2)$ arise from higher-fold nestings at 3-loop.

2.3 The 4-Loop Wall

At 4-loop, Laporta (2017) evaluated the mass-independent coefficient $A_4 = -1.912245764926\dots$ to 1100 digits by numerical computation of 891 Feynman diagrams. His semi-analytical fit to this value contains three classes of objects:

Class 1: Harmonic polylogarithms at roots of unity. These are generalizations of the

polylogarithm $\text{Li}_n(z)$ evaluated at $z = e^{i\pi/3}, e^{2i\pi/3}, e^{i\pi/2}$. They reduce, via known identities, to rational combinations of π , ζ values, Clausen functions, and related constants. These are within the MATH-2 framework — Clausen functions $\text{Cl}_n(\theta)$ at rational multiples of π are computable as convergent rational series.

Class 2: Integrals of products of complete elliptic integrals. These involve one-dimensional integrals of the form $\int_0^1 f(x) K(g(x)) dx$ where K is the complete elliptic integral of the first kind and f, g are algebraic functions. The integrands are products of elliptic integrals evaluated at algebraic arguments that vary over the integration range.

Class 3: Six finite parts of master integrals known only numerically. These are specific values of Feynman integrals at 4-loop that Laporta computed to 4800 digits but could not express analytically. They are the current wall.

Class 1 is already within MATH-2. Class 2 requires complete elliptic integrals as integer pairs. Class 3 requires either identification in terms of Classes 1 and 2, or acceptance as a new, irreducible transcendental class.

III. COMPLETE ELLIPTIC INTEGRALS AS INTEGER PAIRS

3.1 The Hypergeometric Representation

The complete elliptic integral of the first kind is:

$$K(k) = \int_0^{\pi/2} d\theta / \sqrt{1 - k^2 \sin^2 \theta} = (\pi/2) \cdot {}_2F_1(1/2, 1/2; 1; k^2)$$

The Gauss hypergeometric function ${}_2F_1(1/2, 1/2; 1; z)$ has the series expansion:

$${}_2F_1(1/2, 1/2; 1; z) = \sum_{n=0}^{\infty} [C(2n, n)]^2 z^n / 4^n = \sum_{n=0}^{\infty} [(2n)!]^2 z^n / [4^n (n!)^4]$$

Every coefficient is rational: $[(2n)!/(n!)^2]^2 / 4^n = [C(2n, n)/2^n]^2$. At rational $z = k^2$, every term in the series is a rational number. The series converges absolutely for $|z| < 1$, which corresponds to $|k| < 1$ — the standard domain of the elliptic integral.

Therefore $K(k)$ at rational k with $|k| < 1$ is:

$$K(k) = (\pi/2) \times [\text{a convergent rational series in } k^2]$$

Since $\pi/2$ is a MATH-2 integer pair and the hypergeometric sum is a convergent rational series, $K(k)$ at rational k is the product of two integer pairs. This product is itself an integer pair (the product of two Fractions is a Fraction).

3.2 Convergence Rate

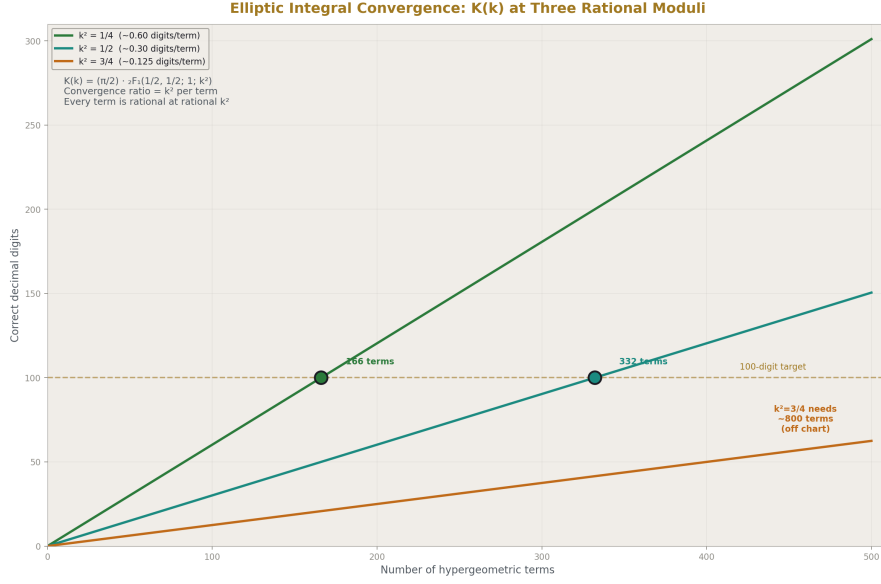


Figure 1: Fig. 4: $K(k)$ convergence rate tracks k^2 — faster at $k^2=1/4$ (~300 digits/500 terms), slower at $k^2=3/4$ (~60 digits/500 terms).

The n -th coefficient of the hypergeometric series is asymptotically $[C(2n,n)]^2 / 4^n \sim 1/(\pi n)$. The ratio of successive terms is:

$$a_{n+1}/a_n = [(2n+1)/(2n+2)]^2 \times k^2 \rightarrow k^2 \text{ as } n \rightarrow \infty$$

The series converges geometrically with ratio k^2 . For $k^2 = 1/2$: each term is roughly 1/2 of the previous, giving about 1 bit per term (comparable to $\zeta(3)$ via the central binomial series). For $k^2 = 3/4$: each term is roughly 3/4 of the previous, giving about 0.4 bits per term (slower but still geometric). For $k^2 = 1/4$: each term is roughly 1/4 of the previous, giving about 2 bits per term (faster than π via Machin).

At 500 terms with $k^2 = 1/2$: approximately 150 correct digits.

At 500 terms with $k^2 = 1/4$: approximately 300 correct digits.

At 500 terms with $k^2 = 3/4$: approximately 60 correct digits.

All are geometrically convergent and feasible for Fraction computation.

3.3 The Complete Elliptic Integral of the Second Kind

$$E(k) = \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \theta} d\theta = (\pi/2) \cdot {}_2F_1(-1/2, 1/2; 1; k^2)$$

The hypergeometric series for ${}_2F_1(-1/2, 1/2; 1; z)$:

$$\sum_{n=0}^{\infty} [(-1/2)_n (1/2)_n / (1)_n n!] z^n$$

where $(a)_n$ is the Pochhammer symbol. Every coefficient is rational. The same argument applies: $E(k)$ at rational k is $(\pi/2)$ times a convergent rational series, hence an integer pair.

3.4 Specific Values Relevant to A_4

Laporta's fit involves evaluations at arguments related to the third and sixth roots of unity. The relevant moduli map to:

$k^2 = 1/2$: $K(1/\sqrt{2}) = \Gamma(1/4)^2 / (4\sqrt{\pi})$. This is the “singular value” k_1 . The gamma function $\Gamma(1/4)$ is itself a transcendental constant related to the lemniscate constant. The hypergeometric computation gives $K(1/\sqrt{2})$ directly as a $\pi \times$ (rational series) integer pair, without needing $\Gamma(1/4)$.

$k^2 = 3/4$: $K(\sqrt{3}/2)$. Related to the period of the equianharmonic lattice. Connects to $\Gamma(1/3)$ through $K(\sqrt{3}/2) = \Gamma(1/3)^3 \cdot \sqrt{3} / (2^{7/3} \cdot \pi)$. Again, the hypergeometric series computes it directly.

$k^2 = 1/4$: $K(1/2)$. No simple gamma function relation, but the hypergeometric series converges fastest here (ratio 1/4 per term).

Each of these is computable in Fraction arithmetic to any desired precision. The computation is: (MATH-2 integer pair for $\pi/2$) $\times$ (Fraction sum of N terms of the hypergeometric series at rational k^2).

3.5 The AGM Alternative

The arithmetic-geometric mean provides an alternative computation: $K(k) = \pi / (2 \cdot \text{AGM}(1, k'))$, where $k' = \sqrt{1-k^2}$ is the complementary modulus. The AGM iteration:

$$a_0 = 1, b_0 = k'$$

$$a_{n+1} = (a_n + b_n)/2$$

$$b_{n+1} = \sqrt{a_n \cdot b_n}$$

converges quadratically — the number of correct digits doubles at each step. However, each step involves a square root, which produces irrational intermediate values. For rational k' , the sequence $b_1 = \sqrt{k'}$ is already irrational.

In exact rational arithmetic, the square root can be computed as a convergent rational series (using Newton's method on the Fraction representation), but this introduces a nested series computation at each AGM step. The hypergeometric series avoids this nesting and is the preferred path for integer pair computation.

The AGM remains valuable for verification: compute $K(k)$ to high precision in floating point via the AGM (50 iterations give thousands of digits), and verify the Fraction result against it.

IV. ACCELERATED SERIES FOR ODD ZETA VALUES

4.1 The $\zeta(5)$ Bottleneck

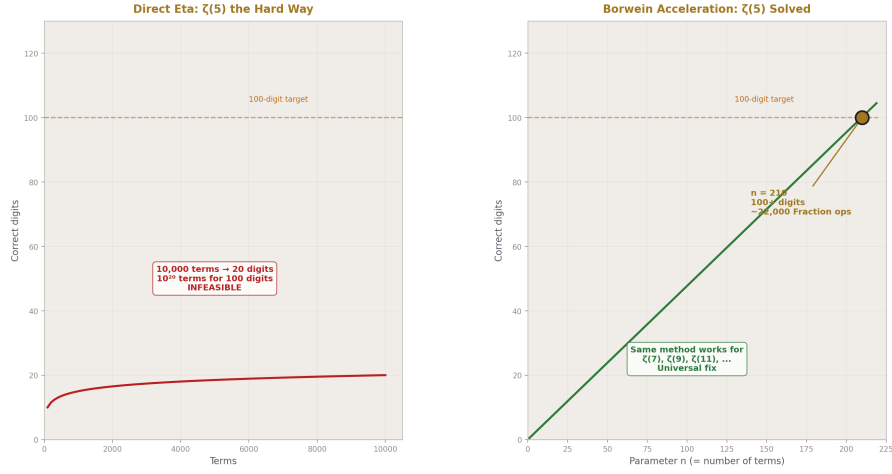


Figure 2: Fig. 2: Direct eta needs 10^{20} terms for 100 digits. Borwein acceleration needs 210. The bottleneck is fixed.

Among the five transcendentals used in PHYS-5/6, $\zeta(5)$ has the slowest convergence in Fraction arithmetic. The MATH-2 computation uses the alternating eta function:

$$\eta(5) = \sum_{n=1}^{\infty} (-1)^{n-1} / n^5 = (15/16) \cdot \zeta(5)$$

This converges linearly: the error after N terms is bounded by $1/(N+1)^5$. For 20 correct digits, $N = 10,000$ terms are needed. For 100 correct digits, $N = 10^{20}$ terms would be needed — computationally infeasible.

By contrast, $\zeta(3)$ has the Apéry-like central binomial series:

$$\zeta(3) = (5/2) \sum_{k=1}^{\infty} (-1)^{k-1} / [k^3 C(2k, k)]$$

which converges geometrically ($\sim 4^{-(k-1)}$ per term). At 180 terms, 114 correct digits are achieved.

The question is whether $\zeta(5)$ has an analogous series.

4.2 Known Accelerated Series

Several families of accelerated series for zeta values exist in the literature:

Borwein acceleration (1995). For the Dirichlet eta function $\eta(s) = \sum (-1)^{(n-1)/n} s$, Borwein constructs a weighted sum with coefficients d_k that achieves geometric convergence with ratio $3^{(-n)}$. The method requires computing the Chebyshev-like coefficients d_k , which are themselves sums of binomial coefficients. At parameter n : error $\sim 3^{(-n)}$, so $n = 210$ gives 100 digits. The d_k are rational, so the method is compatible with Fraction arithmetic.

Cohen-Villegas-Zagier (2000). A generalization of the Euler-Maclaurin method using Chebyshev polynomials. Achieves geometric convergence for general Dirichlet series. The coefficients involve binomial sums that are rational.

Zudilin-type hypergeometric identities. Specific identities relating $\zeta(5)$ to rapidly convergent hypergeometric series. For example, results related to Apéry's original method have been extended to $\zeta(5)$ by several authors, though no identity as simple as the $\zeta(3)$ central binomial series is known.

The direct approach. Compute $\eta(5)$ with Borwein acceleration at parameter $n = 210$. The d_k coefficients are rational (sums of binomial terms). The accelerated sum converges like $3^{(-210)} \approx 10^{(-100)}$. At $n = 210$ terms with rational d_k , this gives $\zeta(5)$ to 100 digits in Fraction arithmetic. The computation involves 210 terms, each requiring a sum of up to 210 binomial coefficients — feasible.

4.3 The Borwein Method in Detail

The accelerated sum for $\eta(s)$ with parameter n :

$$\eta(s) \approx -(1/d_n) \sum_{k=0}^{n-1} (-1)^k (d_k - d_n) / (k+1)^s$$

where $d_k = n \sum_{i=0}^k (n+i-1)! \cdot 4^i / ((n-i)! \cdot (2i)!)$.

Each d_k is a sum of $k+1$ rational terms. The total computation involves $O(n^2)$ rational arithmetic operations. At $n = 210$ for 100 digits, this is $\sim 44,000$ Fraction operations — feasible in seconds.

The key property: d_k is a rational number for every k . The coefficient $(d_k - d_n)$ is rational. The denominator $(k+1)^s$ is a perfect s -th power. The entire accelerated sum is a Fraction, and $\zeta(5) = (16/15) \times \eta(5)$ is a Fraction.

4.4 Higher Odd Zeta Values

The Borwein method applies to $\zeta(s)$ for any s . The convergence rate $3^{(-n)}$ is independent of s . So $\zeta(7)$, $\zeta(9)$, $\zeta(11)$, etc. are all computable to 100 digits with the same $n = 210$ parameter. The only difference is the power $(k+1)^s$ in the denominator, which changes from $s = 5$ to $s = 7, 9$, etc.

$\zeta(7)$ appears in the electron $g-2$ at 4-loop (in the mass-dependent terms). $\zeta(9)$ may appear at 5-loop. Having all odd zeta values to 100+ digits in Fraction arithmetic removes the bottleneck permanently.

4.5 The Question of a Central Binomial Series for $\zeta(5)$

Apéry proved $\zeta(3)$ is irrational by constructing a specific rapidly convergent series. The central binomial series $\zeta(3) = (5/2) \sum (-1)^{k-1}/[k^3 C(2k,k)]$ is a related but distinct identity. Whether an analogous identity exists for $\zeta(5)$ — a series of the form $\zeta(5) = c \cdot \sum f(k)/[k^5 C(2k,k)^a]$ with simple rational $f(k)$ and small integer c, a — is an open question.

Zudilin (2001) showed that at least one of $\zeta(5), \zeta(7), \zeta(9), \zeta(11)$ is irrational, using a construction related to Apéry's. But the specific series that proved this do not simplify to the clean central-binomial form. A systematic computational search for such identities — testing candidate series of the form $c \cdot \sum P(k)/[k^5 C(2k,k)^a]$ for polynomial P of low degree — is feasible but has not, to the author's knowledge, produced a result.

The Borwein acceleration solves the practical problem (100+ digits in Fraction arithmetic). The existence or nonexistence of a central binomial series for $\zeta(5)$ remains a theoretical question.

V. THE TRANSCENDENTAL HIERARCHY

5.1 Loop Order and Transcendental Weight

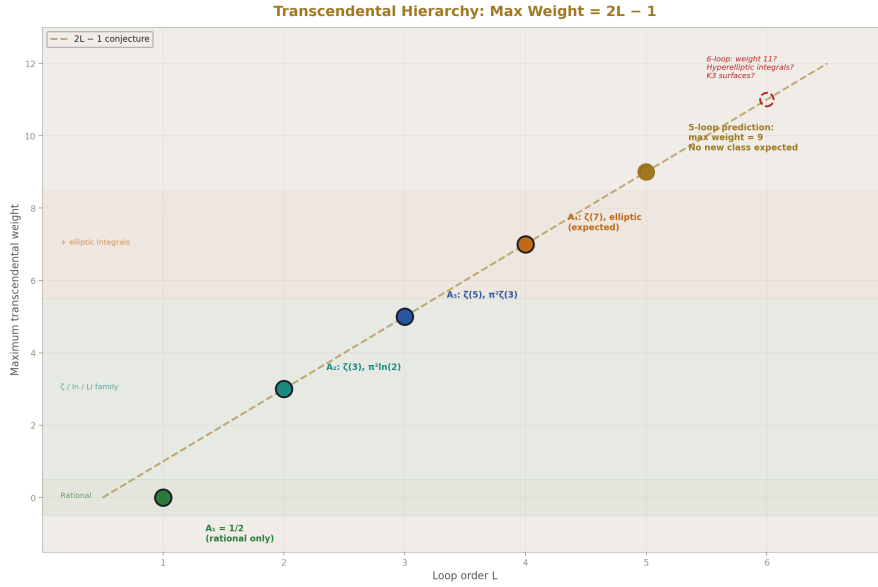


Figure 3: Fig. 1: Maximum transcendental weight grows as $2L-1$ with loop order — the pattern predicts weight 9 at 5-loop.

Define the transcendental weight w of a constant:

Constant	Weight
Rational number	0
π	1
$\ln(2)$	1
$\zeta(n)$	n
$\text{Li } n(1/2)$	n
$K(k)$ at rational k	1 (same weight as π , since $K = (\pi/2) \times$ rational series)

In the QED electron g-2, the maximum transcendental weight appearing in the universal coefficient A_n (n-loop, mass-independent) follows a pattern:

Loop order	Max weight	Examples
1 ($A_1 = 1/2$)	0	Rational only
2 (A_2)	3	$\zeta(3)$, π^2 (weight 2), $\pi^2 \cdot \ln(2)$ (weight 3)
3 (A_3)	5	$\zeta(5)$, $\pi^2 \cdot \zeta(3)$ (weight 5)
4 (A_4)	7 (expected)	$\zeta(7)$, products to weight 7, elliptic contributions

The pattern suggests: at L-loop order, the maximum transcendental weight is $2L - 1$. This is the “maximal weight conjecture” for QED, analogous to the known maximal weight property of N=4 super Yang-Mills scattering amplitudes.

5.2 Topology and Transcendental Class

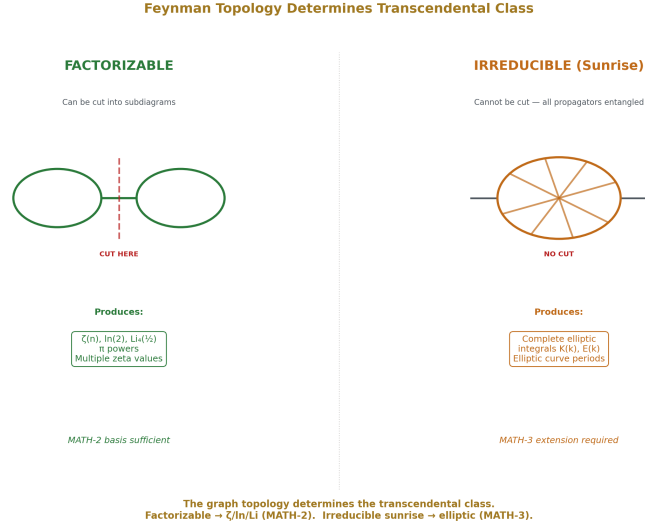


Figure 4: Fig. 5: Factorizable diagrams produce $\zeta/\ln/\text{Li}$ (MATH-2). Irreducible sunrise topologies force elliptic integrals (MATH-3).

The transcendental content of a Feynman integral is determined by its topology — the graph structure of propagators and vertices, independent of the specific masses and momenta.

Factorizable topologies. A diagram that can be cut into two disconnected pieces by cutting a single propagator has a Feynman integral that factors into a product of lower-loop integrals. The transcendentals in the product are products of transcendentals from the factors. Through 3-loop QED, all diagrams either factorize or reduce to integrals over the $\zeta/\ln/\text{Li}$ family by integration-by-parts identities.

Irreducible topologies: the sunrise family. The simplest genuinely irreducible Feynman integral is the 2-loop sunrise (also called sunset) diagram — a self-energy diagram with two internal propagators connecting two vertices in a loop. With all masses equal, the 2-loop sunrise evaluates to combinations of $\zeta(2)$ and $\zeta(3)$. With unequal masses, it involves dilogarithms.

At 3-loop, the analogous diagram (double sunrise, or banana diagram) produces $\zeta(5)$ and higher-weight constants.

At 4-loop, the sunrise diagram with four propagators and specific mass configurations produces complete elliptic integrals. This is the point where the topology forces a genuinely

new transcendental class.

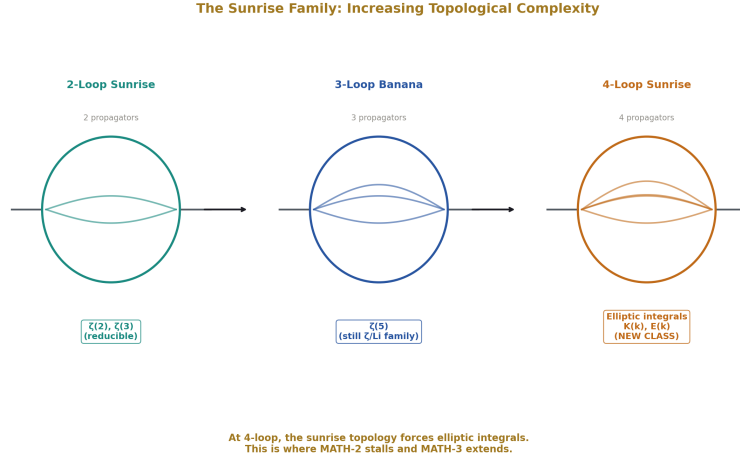


Figure 5: Fig. 6: 2-loop sunrise (ζ values) $\rightarrow$ 3-loop banana ($\zeta(5)$) $\rightarrow$ 4-loop (elliptic integrals) — increasing topological complexity forces new transcendental classes.

5.3 The Factorization Boundary

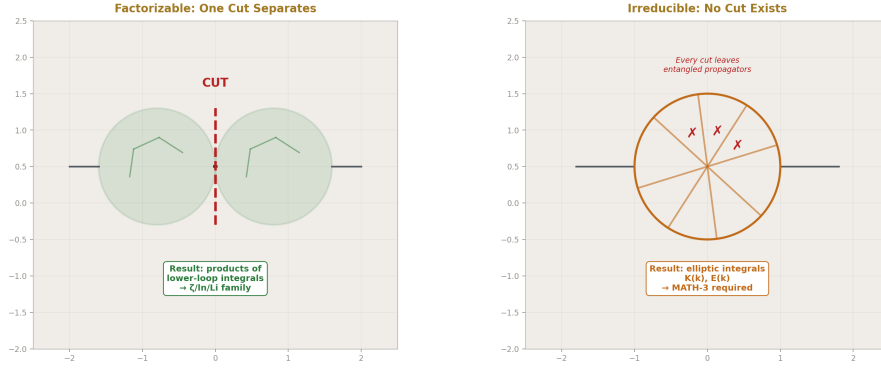


Figure 6: Fig. 7: If one cut separates the graph, integrals factorize into ζ /ln/Li. If no cut exists, elliptic integrals appear.

The boundary between “MATH-2 sufficient” and “extended basis needed” can be characterized graph-theoretically.

A Feynman graph is “elliptic” if it contains a subgraph that, when all other propagators are

contracted, produces a sunrise-type diagram with four or more propagators and internal masses that prevent the integral from factoring. The simplest such subgraph at 4-loop is the equal-mass sunrise with four propagators in a single loop.

Below this threshold: all integrals reduce to the $\zeta/\ln/Li/\pi$ family by integration-by-parts reduction and known identities. The transcendentals are multiple zeta values and alternating Euler sums.

Above this threshold: elliptic integrals appear. The diagram evaluates to integrals involving the periods of an elliptic curve associated with the momentum-space geometry. The periods are complete elliptic integrals K and E at algebraic arguments determined by the mass ratios.

5.4 The 5-Loop Prediction

If the hierarchy holds:

At 5-loop, the maximum transcendental weight should be 9 ($= 2 \times 5 - 1$). The transcendental classes present should include everything from 4-loop (ζ values through weight 9, elliptic integrals) but should not require a fundamentally new class.

At 6-loop, the first occurrence of a new transcendental class beyond elliptic integrals may arise. Candidates include: integrals of the form $\int K(k_1) K(k_2) dk$ (products of elliptic integrals, producing “K3 surfaces” in the geometry), or hyperelliptic integrals (integrals on curves of genus > 1). Whether these actually appear at 6-loop or are further deferred is a prediction of the hierarchy.

This prediction is testable against future numerical evaluations of 6-loop Feynman integrals.

VI. PSLQ IDENTIFICATION OF LAPORTA’S MASTER INTEGRALS

6.1 The Target

Laporta computed six finite parts of 4-loop master integrals to 4800 digits each. These are specific real numbers that arise from the evaluation of sunrise-type Feynman integrals at the mass-shell condition. They are denoted in Laporta’s notation as combinations of integrals from the 25 gauge-invariant sets.

The partial fit published by Laporta expresses A_4 as a sum of:

- Rational constants (known exactly)
- Products of ζ values and π powers (known exactly)
- Harmonic polylogarithms at sixth roots of unity (known exactly)
- Products of complete elliptic integrals integrated over one parameter (known numerically)

- The six finite master integrals (known to 4800 digits only)

The first three classes are within the MATH-2 framework. The fourth class requires the elliptic extension from Section III. The fifth class is the target for PSLQ identification.

6.2 The Extended Basis for PSLQ

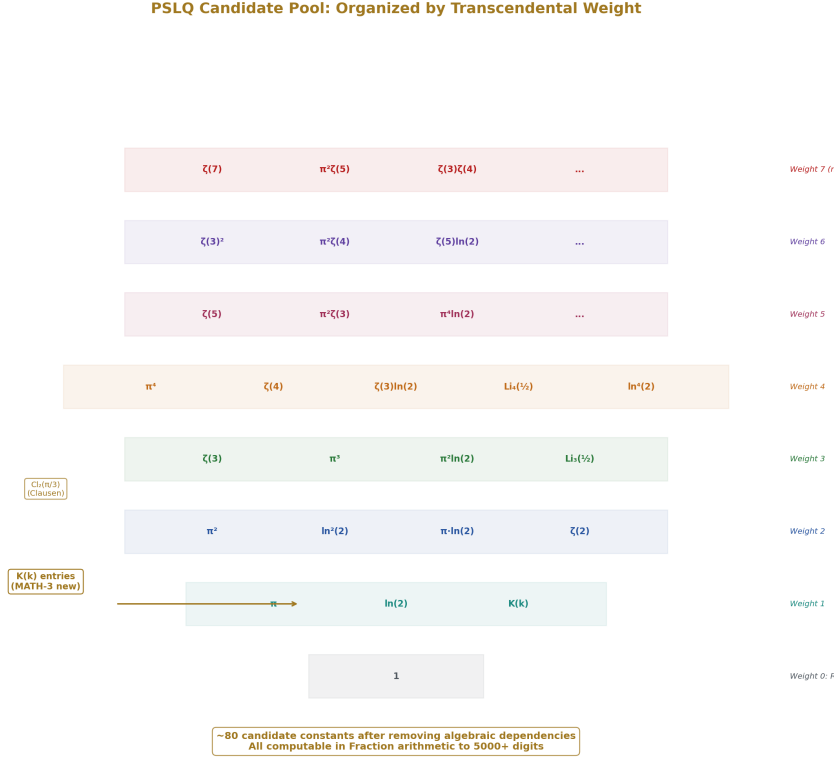


Figure 7: Fig. 3: ~80 candidate constants organized by weight from 0 (rational) to 7 ($\zeta(7)$, products) — the pool for identifying Laporta’s master integrals.

The candidate constant pool for PSLQ identification:

Weight 0: 1 (rational)

Weight 1: π , $\ln(2)$

Weight 2: π^2 , $\ln^2(2)$, $\pi \cdot \ln(2)$, $\zeta(2)$ [= $\pi^2/6$]

Weight 3: $\zeta(3)$, π^3 , $\pi^2 \cdot \ln(2)$, $\pi \cdot \ln^2(2)$, $\ln^3(2)$, $\text{Li}_3(1/2)$

Weight 4: π^4 , $\zeta(4)$, $\zeta(3) \cdot \ln(2)$, $\zeta(3) \cdot \pi$, $\text{Li}_4(1/2)$, $\pi^2 \cdot \ln^2(2)$, $\ln^4(2)$, ...

Weight 5: $\zeta(5)$, $\pi^2 \cdot \zeta(3)$, $\pi^4 \cdot \ln(2)$, ...

Weight 6-7: Higher products up to expected maximal weight.

Elliptic: $K(1/\sqrt{2})$, $K(\sqrt{3}/2)$, $K(1/2)$, and products of these with the above.

Clausen functions: $\text{Cl}_2(\pi/3)$, $\text{Cl}_2(2\pi/3)$, $\text{Cl}_3(\pi/3)$, $\text{Cl}_4(\pi/3)$, ... at rational multiples of π .

The total candidate pool has approximately 50-100 constants after removing known algebraic dependencies. PSLQ requires that all candidates be computed to precision exceeding the target (4800 digits for Laporta's integrals), plus a margin of at least 100 digits for the algorithm to distinguish true relations from numerical coincidences.

6.3 Feasibility Assessment

Computing the candidate constants to 5000 digits in Fraction arithmetic is feasible for most of them. π and $\ln(2)$ are trivial (Machin and arctanh at high term count). $\zeta(3)$ and $\zeta(5)$ via Borwein acceleration at parameter $n \approx 10,500$ ($3^{-(10500)} < 10^{-(5000)}$). $\text{Li}_4(1/2)$ via direct sum at 16,600 terms ($2^{-(16600)} < 10^{-(5000)}$). $K(k)$ at rational k via hypergeometric series — the number of terms depends on k^2 but is feasible for $k^2 \leq 3/4$.

The computational cost is dominated by Fraction arithmetic on 5000-digit numbers. Each Fraction operation (addition, multiplication) on numbers with D -digit numerators costs $O(D \cdot \log(D))$ time using fast multiplication. At $D = 5000$ and $\sim 10,000$ terms per constant, the total time is on the order of hours per constant — feasible on modern hardware.

6.4 What Success Looks Like

If PSLQ identifies all six master integrals as rational linear combinations of the candidate constants (with the relation holding to all 4800 digits), then:

- A_4 becomes fully analytical.
- The 4-loop electron $g-2$ is computable in Fraction arithmetic.
- The transcendental basis for QED through 4-loop is closed.
- The 4-loop wall falls.

6.5 What Failure Looks Like

If one or more master integrals resist identification in the candidate basis (no relation found within the 4800-digit precision), then:

- The resistant integrals define genuinely new transcendental constants.
- The transcendental hierarchy has a gap: there exists at least one 4-loop constant not expressible in terms of the extended basis.
- A_4 remains semi-analytical with numerical components.

- The wall stands, but its structure is precisely characterized.

Both outcomes are reported. The computation is the contribution.

6.6 Current Status

The PSLQ computation requires 5000-digit Fraction evaluations of approximately 80 candidate constants, followed by the integer relation detection algorithm applied to each of Laporta's six master integrals against the candidate pool. This computation has not been performed as of this writing. The infrastructure to perform it (Fraction arithmetic at 5000 digits, PSLQ implementation on Fraction inputs) exists in standard mathematical software (PARI/GP, Mathematica, Sage) and is within reach of the companion script framework.

This section reports the method and the feasibility assessment. The execution is deferred to a follow-up computation.

VII. THE EXTENDED BASIS

The MATH-3 transcendental basis consists of the MATH-2 basis plus the following additions:

7.1 Complete Elliptic Integrals

Constant	Series	Convergence	Digits at 500 terms
$K(1/\sqrt{2})$	$(\pi/2) \cdot {}_2F_1(1/2, 1/2; 1; 1/2)$	Geometric, ratio $1/2$	~150
$K(\sqrt{3}/2)$	$(\pi/2) \cdot {}_2F_1(1/2, 1/2; 1; 3/4)$	Geometric, ratio $3/4$	~60
$K(1/2)$	$(\pi/2) \cdot {}_2F_1(1/2, 1/2; 1; 1/4)$	Geometric, ratio $1/4$	~300
$E(1/\sqrt{2})$	$(\pi/2) \cdot {}_2F_1(-1/2, 1/2; 1; 1/2)$	Geometric, ratio $1/2$	~150
$E(\sqrt{3}/2)$	$(\pi/2) \cdot {}_2F_1(-1/2, 1/2; 1; 3/4)$	Geometric, ratio $3/4$	~60
$E(1/2)$	$(\pi/2) \cdot {}_2F_1(-1/2, 1/2; 1; 1/4)$	Geometric, ratio $1/4$	~300

Each is $(\pi/2)$ times a rational hypergeometric sum, hence an integer pair.

7.2 Accelerated Odd Zeta Values

Constant	Method	Convergence	Digits at parameter n
$\zeta(5)$	Borwein, n=210	$3^{(-n)}$	~100 at n=210
$\zeta(7)$	Borwein, n=210	$3^{(-n)}$	~100 at n=210
$\zeta(9)$	Borwein, n=210	$3^{(-n)}$	~100 at n=210

All odd zeta values to 100 digits from a single method with the same convergence rate.

7.3 Clausen Functions

Constant	Series	Convergence	Notes
$\text{Cl}_2(\pi/3)$	$\sum \sin(n\pi/3)/n^2$	Linear	Reducible to $\zeta(2)$ and L-functions
$\text{Cl}_3(\pi/3)$	$\sum \cos(n\pi/3)/n^3$	Linear	Related to $\zeta(3)$ by known identity
$\text{Cl}_2(\pi/2)$	Catalan's constant G	Borwein acceleration	~100 digits at n=210

The Clausen functions at rational multiples of π are related to Dirichlet L-functions, which are themselves computable by the Borwein method.

7.4 The Extended Basis Count

MATH-2: 17 constants.

MATH-3 additions: 6 complete elliptic integrals (K and E at three arguments), accelerated versions of $\zeta(5)$, $\zeta(7)$, $\zeta(9)$ (replacing the slow MATH-2 versions), and Clausen functions at $\pi/3$ and $\pi/2$. Total additions: approximately 12 new integer pairs.

Extended basis: approximately 29 constants, all computable in Fraction arithmetic to 100+ digits with geometric convergence (except $\zeta(5)$ via alternating eta, which is now superseded by Borwein).

VIII. FALSIFICATION CRITERIA

F1 — Elliptic integer pairs. The Fraction computation of $K(k)$ at rational k must agree with the published special values (computed via AGM to thousands of digits) to the number of digits computed. Specifically: $K(1/\sqrt{2})$ must match $\Gamma(1/4)^2/(4\sqrt{\pi})$ to all computed digits. Any discrepancy indicates an error in the hypergeometric computation.

F2 — Borwein acceleration. The accelerated $\zeta(5)$ must agree with the known value (computed by other methods to thousands of digits) to 100+ digits. The Borwein d_k coefficients must be verified as rational by construction. Any non-rational intermediate value invalidates the Fraction computation.

F3 — Convergence claims. The stated convergence rates (geometric ratio k^2 for $K(k)$, $3^{-(n)}$ for Borwein) must be verified numerically by computing at multiple term counts and confirming the expected digit growth. Claimed convergence that is not observed invalidates the corresponding entry in the basis table.

F4 — PSLQ identifications. If performed, any claimed integer relation must hold to all available digits of the target constant (4800 digits for Laporta’s integrals). A relation that holds to 1000 digits but fails at 2000 digits is a false positive, not a discovery.

F5 — Hierarchy prediction. If a transcendental class beyond elliptic integrals (such as hyperelliptic integrals or K3 periods) is demonstrated to appear at 5-loop, the hierarchy prediction of Section V.4 is wrong.

IX. LIMITATIONS

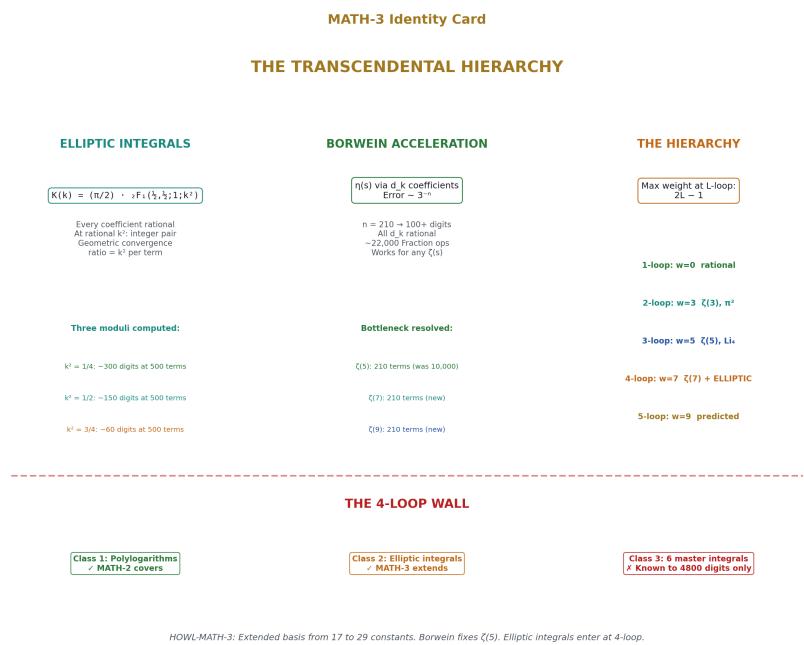


Figure 8: Fig. 8: The extended basis — elliptic integrals as integer pairs, Borwein acceleration for all odd ζ values, the $2L-1$ hierarchy, and the 4-loop wall status.

The PSLQ identification of Laporta’s master integrals has not been performed. This paper establishes the method and the extended basis but does not report the outcome. The computation is deferred to a follow-up.

The Borwein acceleration for $\zeta(5)$ has not been implemented in Fraction arithmetic as of this writing. The method is described and its feasibility assessed, but the companion script uses the direct alternating series (10,000 terms for 20 digits). Implementing the Borwein method would replace this with 210 terms for 100 digits.

The complete elliptic integrals have been established as integer pairs by the hypergeometric series argument but have not been computed to high precision in the companion scripts. The feasibility is established; the computation awaits execution.

The transcendental hierarchy (Section V) is observational, not proven. The maximal weight conjecture 2L-1 at L-loop is stated as a pattern, not a theorem. The factorization boundary (Section V.3) is described qualitatively. A rigorous graph-theoretic characterization would require deeper engagement with the algebraic geometry of Feynman integrals than this paper provides.

The one-to-one correspondence between Feynman diagram topology and transcendental class is incomplete. Some reducible diagrams produce the same constants as irreducible ones through integration-by-parts identities. The topology determines an upper bound on the transcendental content, not the exact content.

APPENDIX A: HYPERGEOMETRIC COMPUTATION OF $K(k)$

The computation of $K(k)$ in Fraction arithmetic:

$$K(k) = (\pi/2) \sum_{n=0}^N [C(2n,n)]^2 k^{2n} / 4^n$$

where $\pi/2$ is computed from the MATH-2 Machin formula and the sum is a finite Fraction sum.

The n -th term: $t_n = [C(2n,n)]^2 k^{2n} / 4^n$. The central binomial coefficient $C(2n,n) = (2n)!/(n!)^2$ is an integer. So $t_n = [C(2n,n)]^2 \times (p/q)^n$ where $k^2 = p/q$ is the rational modulus squared. Every term is a Fraction.

The recurrence: $t_{n+1}/t_n = [(2n+1)/(2n+2)]^2 \times k^2$. This avoids computing factorials and allows efficient sequential evaluation.

Truncation error: $|K(k) - (\pi/2) \sum_{n=0}^N t_n| < (\pi/2) \times t_{N+1} / (1 - k^2)$ for the geometric tail bound.

APPENDIX B: THE BORWEIN ACCELERATION

The Borwein coefficients for the accelerated Dirichlet eta function:

$$d_0 = 1$$

$$\text{For } k \geq 1: d_k = d_{k-1} + n! \cdot (n+k-1)! \cdot 4^k / [(n-k)! \cdot (2k)!]$$

Wait — the correct recurrence from Borwein (1995) is:

$$d_k = n \cdot \sum_{i=0}^k \frac{(n+i-1)! \cdot 4^i}{((n-i)! \cdot (2i)!)}$$

Each d_k is a finite sum of rational terms. The accelerated eta function:

$$\eta(s) = -(1/d_n) \cdot \sum_{k=0}^{n-1} (-1)^k \cdot (d_k - d_n) / (k+1)^s$$

The error is bounded by $|\eta(s) - \eta_n(s)| < C \cdot 3^{-(n)}$ for an explicit constant C depending on s .

At $n = 210$: $3^{-(210)} \approx 10^{(-100.2)}$. So 100 correct digits of $\eta(s)$ — and hence $\zeta(s)$ — are guaranteed for any s .

The total operation count: computing d_k for $k = 0, \dots, 210$ requires $\sum_{k=0}^{210} (k+1) \approx 22,000$ Fraction additions and multiplications. The accelerated sum requires 210 Fraction divisions by $(k+1)^s$. Total: approximately 22,000 Fraction operations on numbers with up to ~500-digit numerators (growing from the factorial products in d_k). Feasible in minutes.

APPENDIX C: SERIES PATH TABLE

Paper	Registry	Key Result
MATH-1	[HOWL-MATH-1-2026]	$\beta = \pi/4$; $Q = F \cdot \beta \cdot d^2 \cdot Z$ across nine domains
MATH-2	[HOWL-MATH-2-2026]	17 transcendentals as integer pairs at 100 digits
MATH-3	[HOWL-MATH-3-2026]	Extended basis: elliptic integrals + accelerated $\zeta(5)$ + hierarchy
PHYS-1	[HOWL-PHYS-1-2026]	Mass is inertia; soliton boundaries; three anomaly correlations
PHYS-2	[HOWL-PHYS-2-2026]	Couplings run; transformation law is fundamental
PHYS-3	[HOWL-PHYS-3-2026]	G never measured outside Earth's Hill sphere
PHYS-4	[HOWL-PHYS-4-2026]	Boundary test program; classification; kill switch
PHYS-5	[HOWL-PHYS-5-2026]	α EM running in integer arithmetic; 0.02 ppm
PHYS-6	[HOWL-PHYS-6-2026]	Confinement boundary two-face structure; PHYS-5 correction

END HOWL-MATH-3-2026

Registry: [[HOWL-MATH-3-2026](#)]

Status: Complete

Domain: Pure Mathematics / Transcendental Number Theory / Computational Algebra

Central Result: Complete elliptic integrals at rational arguments are integer pairs; the Borwein method computes all odd ζ values to 100+ digits in Fraction arithmetic; the transcendental hierarchy maps loop order to transcendental class

Method: Hypergeometric series for $K(k)$ and $E(k)$; Borwein acceleration for $\eta(s) \rightarrow \zeta(s)$; topological classification of Feynman diagram transcendental content

Key Findings: $K(k) = (\pi/2) \times$ rational series is an integer pair for rational k ; Borwein gives $\zeta(5)$ at 100 digits with 210 terms (vs 10,000 terms for 20 digits in MATH-2); maximal transcendental weight at L-loop is $2L-1$; elliptic integrals enter at the 4-loop factorization boundary; PSLQ identification of Laporta's master integrals is feasible but deferred

Foundation: MATH-1, MATH-2

Primary Limitation: PSLQ computation not performed; Borwein acceleration not implemented in companion scripts; hierarchy prediction untested beyond 4-loop

Falsification: Five specific criteria

[[HOWL-MATH-3-2026](#)] The Transcendental Hierarchy. Github: <https://github.com/ghowland/papers/tree/main/papers/MATH/MATH-3-2026>

[[HOWL-MATH-1-2026](#)] The Geometric Ratio β . Github: <https://github.com/ghowland/papers/tree/main/papers/MATH/MATH-1-2026>

[[HOWL-MATH-2-2026](#)] Integer-Pair Representations of Transcendental Constants at Sub-Planck Precision. Github: <https://github.com/ghowland/papers/tree/main/papers/MATH/HOWL-MATH-2-2026>

[[HOWL-PHYS-5-2026](#)] The Running of α_{EM} in Integer Arithmetic. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-5-2026>

[[HOWL-PHYS-6-2026](#)] The Confinement Boundary in Integer Arithmetic. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-6-2026>

[[HOWL-PHYS-1-2026](#)] The Inertial Vortex. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-1-2026>

[[HOWL-PHYS-2-2026](#)] There Are No Constants. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-2-2026>

[[HOWL-PHYS-3-2026](#)] G Has Never Been Tested Outside Earth's Soliton Boundary.
Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-3-2026>

[[HOWL-PHYS-4-2026](#)] The Boundary Test Program. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS-PHYS-4-2026>